Computer Vision

Lecture 3

Segmentation & Visual Saliency

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Segmentation

- Active Contour
- Watershed
- GraphCut
- SLIC SuperPixel

Active Contour

- It is a common type of segmentation algorithm in computer vision.
- It works on the principle of the "snake," where a snake exists on the image and continuously moves to enclose the objects in the image (regions with color variation) with the minimal possible path.
- It works on the principle of minimizing energy, i.e., it measures an energy value - similar to computing the difference between a pixel and its neighbors- and moves inward to make this difference as small as possible.





Active Contour

- Sometimes, a binary image can be provided representing points on which the algorithm should not operate, in order to reduce the required processing time.





Active Contour

- This algorithm aims to minimize the following equation over time.

$$E_{total} = E_{image} + E_{contour}$$

- The computational process mainly depends on differences between a pixel and its neighbors in the image.
- We take uniform samples from the contour boundary.
- We move each sample (point) within a window **W** so that the equation result becomes as small as possible.
- We then sum the values for all samples; if the result becomes smaller than a certain threshold, we stop.

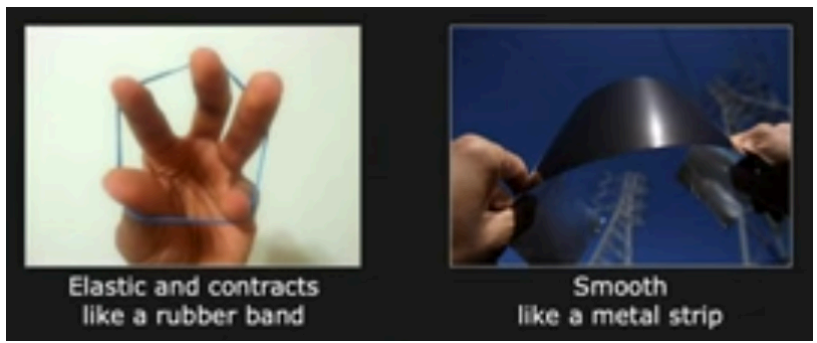


Active Contour

- We calculate the energy of contour

$$E_{contour} = \alpha E_{elastic} + \beta E_{smooth}$$

- where α and β are values provided by the users.





Active Contour

Curve Energy Calculation

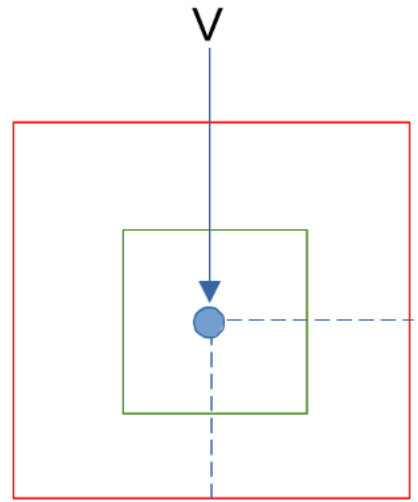
$$E_{contour} = \alpha E_{elastic} + \beta E_{smooth}$$

$$E_{elastic} = \left\| \frac{dv}{ds} \right\|^2 \approx \|v_{i+1} - v_i\|^2 = (x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2$$

$$E_{smooth} = \left\| \frac{d^2v}{ds^2} \right\|^2$$

$$E_{smooth} \approx \|v_{i+1} - 2v_i + v_{i-1}\|^2$$

$$E_{smooth} \approx (x_{i+1} - 2x_i + x_{i-1})^2 + (y_{i+1} - 2y_i + y_{i-1})^2$$



- **S** represents movements within the window surrounding the sample.
- **V**, **α**, and **β** are constants specified by the user.

Active Contour

Image Energy Calculation

- Image energy is based on computing the derivatives of illumination between neighboring pixels.

$$E_{image} = \|\Delta I(x, y)\|^2$$

- Total Energy

$$E_{total} = E_{image} + \alpha E_{elastic} + \beta E_{smooth}$$

- Image energy is called external energy and pulls the curves toward edges, while curve energy is called internal energy, and its components:
  - $E_{elastic}$ Force the curve to remain continuous and contracted
  - E_{smooth} Force the curve to remain smooth

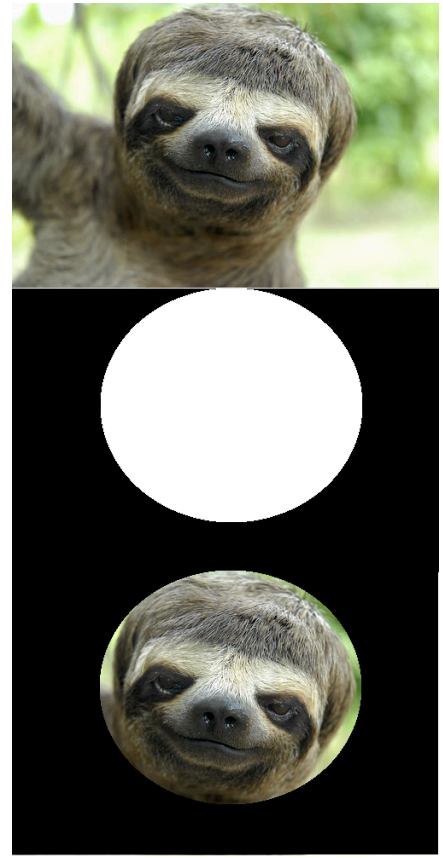
Active Contour

- Active contours are useful for tracking objects in video frames.
- In the first row of images, the object can be tracked despite contour deformation between frames due to motion.
- In the second row, the object can be tracked despite contour deformation caused by change of viewpoint.



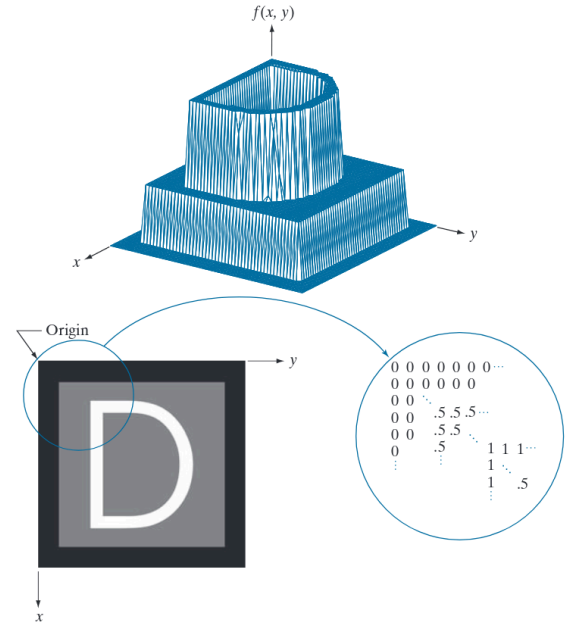
Active Contour

- To speed up this algorithm, we often provide a simple **mask** to avoid operating on the entire image.
- Only the object's region is processed, reducing execution time.



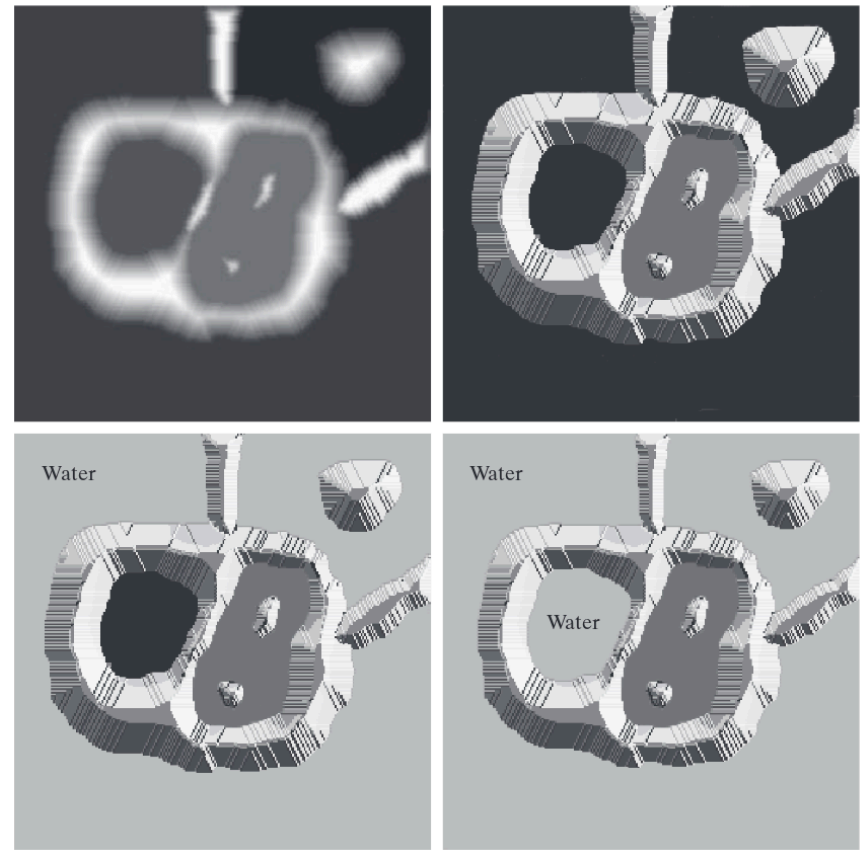
Watershed

- This algorithm treats the grayscale image as a topographic map.
- The higher the grayscale value, the higher the mountain peak.
- In this topographic representation, pixels are classified into three types:
  1. Points that represent **local minima**
  2. Points where, if a drop of water falls, it flows toward a local minimum
  3. Points from which water can flow toward more than one local minimum
- Points satisfying condition (2) represent the **catchment basin**.
- Points satisfying condition (3) represent the **watershed lines**.
- The algorithm aims to find these watershed lines.



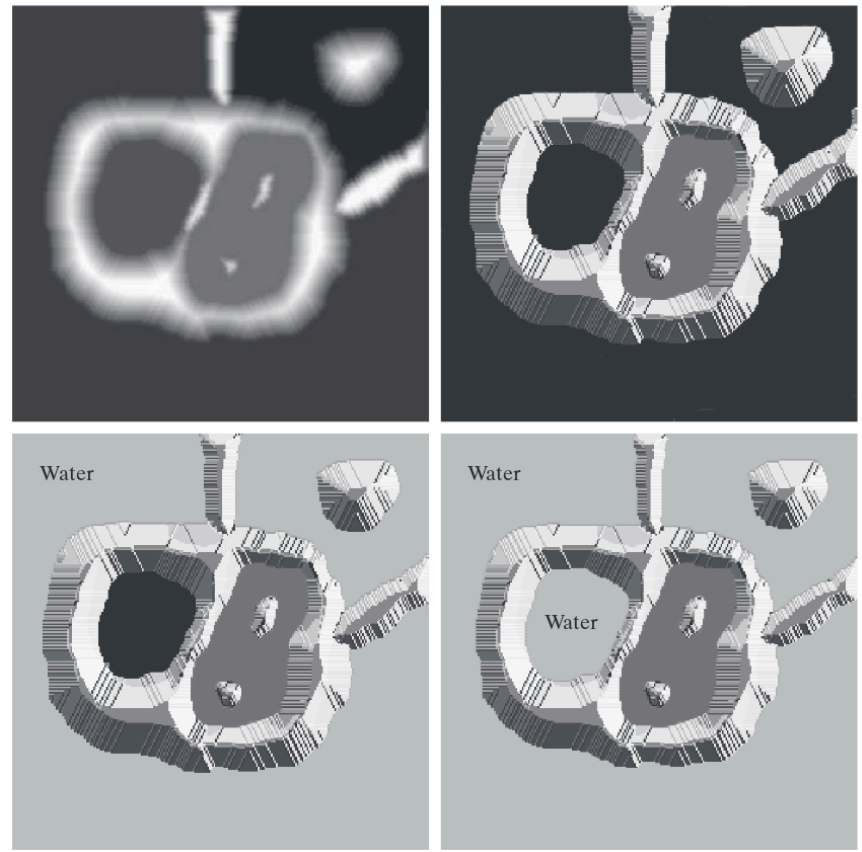
Watershed

- Given the following images: the upper-left image is the original image, and the upper-right is the topography.



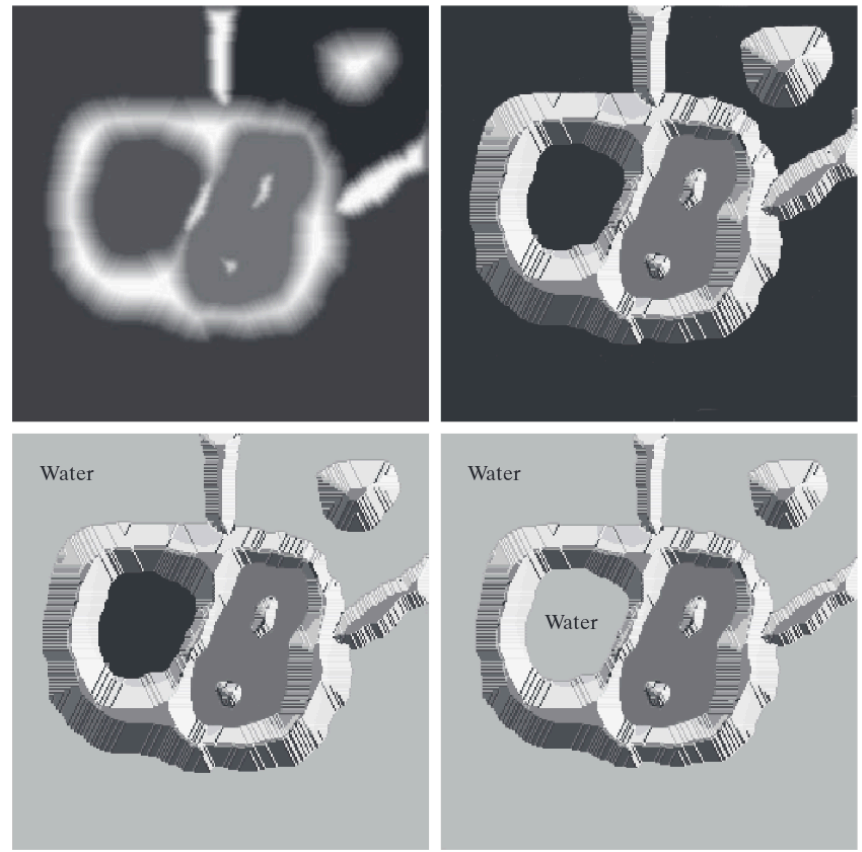
Watershed

- We surround the image with dams - structures with a height equal to the maximum possible white level.
- We start filling the image with water.
- The background fills first, then the left basin.



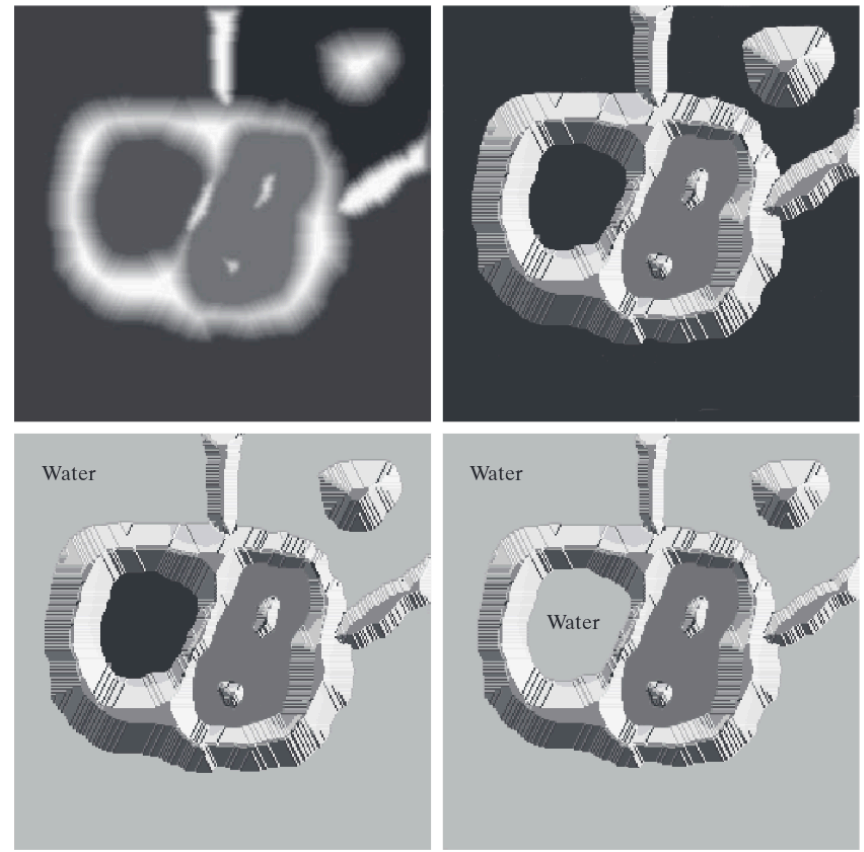
Watershed

- Then the right basin fills.
- We continue until reaching the upper-right image.
- Here the boundary between the right and left basins becomes one pixel wide (an edge – condition 3).
- Here, we build a dam over this edge to prevent water from crossing.



Watershed

- As filling continues, edges shrink to one-pixel thickness, and we build dams at each edge.
- Finally, we obtain the lower-right image.



Watershed

- One of the most important applications of Watershed is extracting semi-regular objects from images.
- To speed up the algorithm, it is often applied to the **image derivative** rather than the image itself, because differences are more visible in the derivative.
- Most suitable is the second derivative as it produces thinner edges

Graph Cut (GraphCuts)

- Graph Cut is an optimization technique widely used in **computer vision** to solve tasks such as:
  - Image segmentation
  - Stereo matching
  - Object/background separation
  - Energy minimization problems in general
- It transforms a vision problem into a **graph problem**, then finds the best solution by computing a **minimum cut** on that graph.

Graph Cut (GraphCuts)

What is a Graph Cut?

- A **graph cut** divides a graph into two disjoint sets by cutting edges.
- Each cut has a **cost**, which is the sum of the weights of the edges that are cut.
- In computer vision, we construct a graph where:
  - **Nodes** = pixels (or superpixels)
  - **Edges** = relationships between pixels
  - **Weights** = cost (penalty) for cutting between pixels
- We then find the **minimum cut** that best separates the graph into two sets:
  - **Source (S)** → often “foreground”

    - **Sink (T)** → often “background”
- The algorithm finds the partition of nodes that minimizes the overall cost.

Graph Cut (GraphCuts)

- GraphCut solves an energy minimization problem of the form:

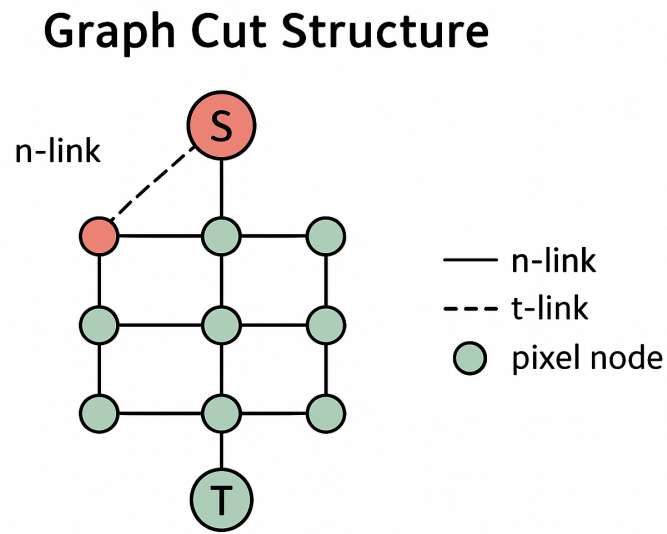
$$E = E_{data} + E_{smooth}$$

- Where:
  - **E_data**: cost of assigning pixel to foreground/background
  - **E_smooth**: cost of assigning *neighboring pixels* different labels (encourages smooth boundaries)
- Graph Cut finds the labeling that minimizes **E**.

Graph Cut (GraphCuts)

How the Graph is Constructed

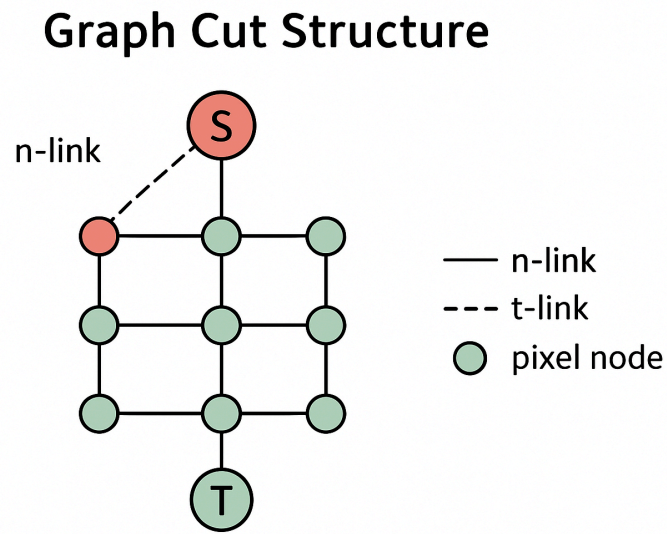
- We build a **directed graph** with:
  1. **Pixel nodes**
     - Each pixel = 1 node.
  2. Two extra nodes:
     - **Source (S)** → represents foreground
     - **Sink (T)** → represents background



Graph Cut (GraphCuts)

How the Graph is Constructed

- Two types of edges:
  1. **t-links (terminal links)**
     - Connect each pixel to S or T.
     - Weights represent the data term:
       - Cost of assigning pixel to foreground (S)
       - Cost of assigning pixel to background (T)
  2. n-links (neighborhood links)
     - Edges between neighboring pixels.
     - Weights represent the smoothness term:
       - High weight = don't cut this edge (keeps region smooth)
       - Low weight = edge can be cut

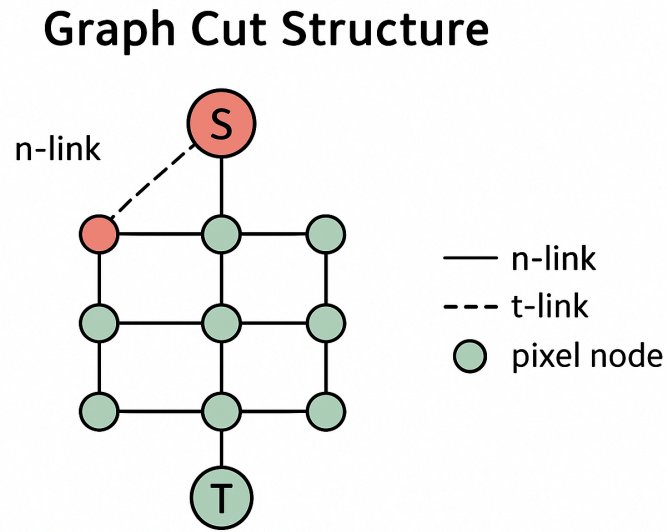


Graph Cut (GraphCuts)

- Once Graph is Built → Find the Minimum Cut
- We run a **max-flow / min-cut algorithm** (like Boykov–Kolmogorov).
- This gives:
  - A partition into two sets (Source side vs Sink side)
  - A boundary where edges were cut → object boundary
- This is the final segmentation result.

Graph Cut (GraphCuts)

- Example: Simple Foreground/Background Segmentation
  - Imagine each pixel has likelihood:
    - More red → belongs to object
    - More green → belongs to background
  - We set:
    - t-links to source: high weight for red pixels
    - t-links to sink: high weight for green pixels
  - GraphCut finds a boundary where:
    - Object region = connected to **Source**
    - Background = connected to **Sink**
  - Smoothness term prevents noisy zig-zag boundaries.



Graph Cut (GraphCuts)

The Energy Function Used by GraphCut

- Typically:

$$E(L) = \Sigma_i D_i(L_i) + \Sigma_{i,j} V_{i,j}(L_i, L_j)$$

- Where:
  - L_i: label (0 = background, 1 = foreground)
  - **Data term** D_i: cost of assigning pixel i a label
  - **Smoothness term** $V_{i,j}$: penalty for neighboring labels being different

Graph Cut (GraphCuts)

Why ?

- Global optimization
- Fast on large images
- Robust to noise
- Handles complex energy functions
- Works with user scribbles (GrabCut)

Uses

- Used in:
  - Medical imaging
  - Photo editors (Photoshop uses it!)
  - Stereo vision
  - Object extraction

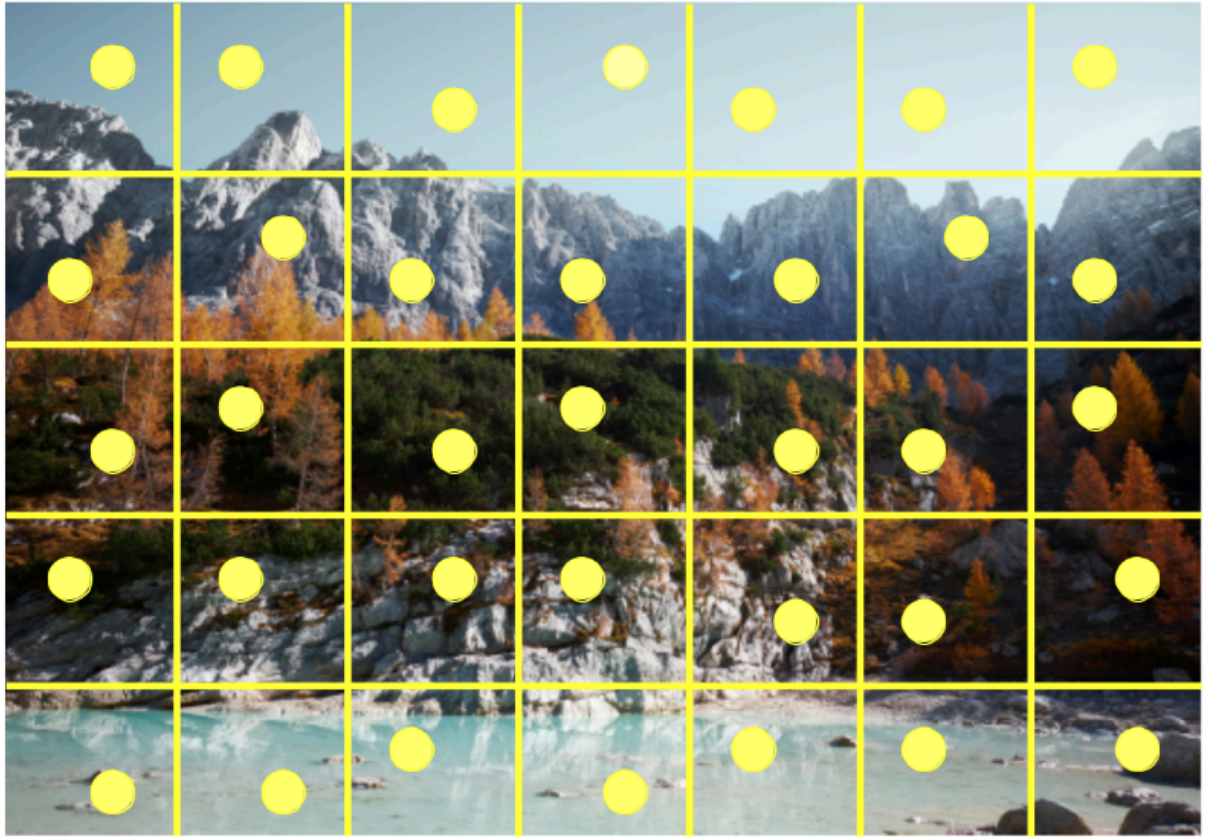
SLIC SuperPixel Algorithm

- This algorithm belongs to the family of segmentation algorithms that detect **superpixels**.
- A superpixel is considered the true shape of the pixel that should have been obtained during imaging.
- It is one of the most used superpixel algorithms due to its speed and result quality.

SLIC SuperPixel Algorithm

- This algorithm divides the image into squares representing initial
- superpixels by drawing a grid of size **S × S**, $S = \sqrt{\frac{N}{K}}$ where:
  - **N** is the total number of pixels
  - **K** is the desired number of superpixels
- The initial superpixel centers are chosen as the smallest gradient value within a 3×3 neighborhood inside each square.

SLIC SuperPixel Algorithm

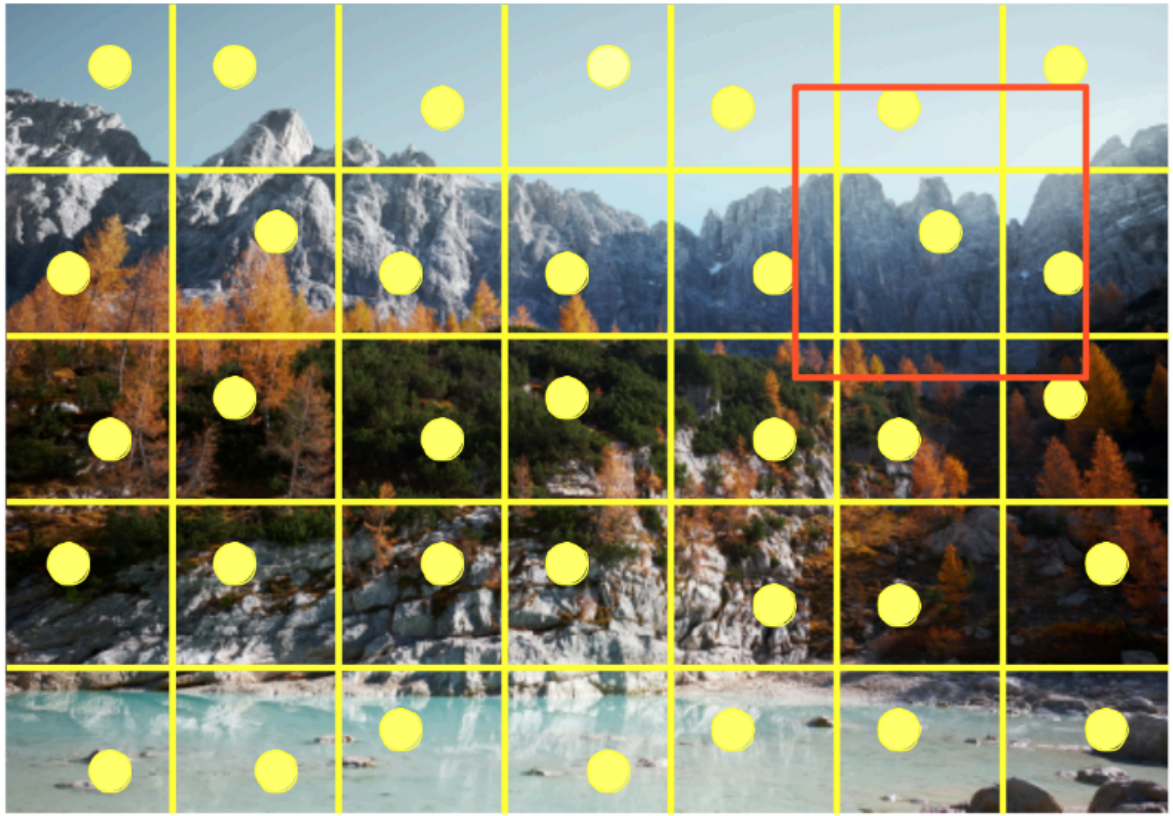


SLIC SuperPixel Algorithm

- We apply **k-means** to each center independently using a neighborhood of size **2S × 2S**.
- We stop when the error becomes smaller than a specified threshold.

SLIC SuperPixel Algorithm

- We apply k-means inside the square



SLIC SuperPixel Algorithm

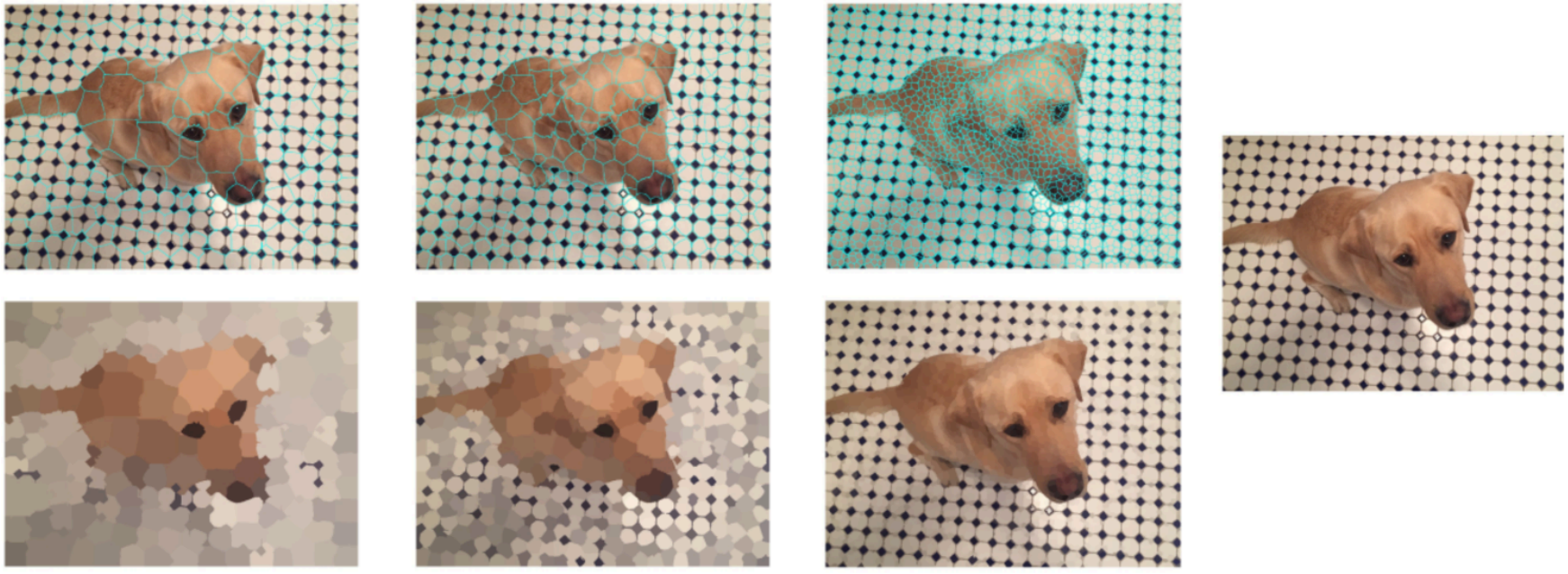
Distance Metric in SLIC

- The SLIC distance depends on color and spatial differences:
  - **d_c**: Euclidean color distance
  - **d_s**: Euclidean spatial distance
- A large **c** value prioritizes spatial distance over color distance, causing superpixels to become square-like at high values.

$$d = \sqrt{d_c^2 + \frac{d_s^2}{s} c^2}$$

SLIC SuperPixel Algorithm

Example



SLIC SuperPixel Algorithm

Purpose of SLIC

- We often need to apply very complex mathematical operations to images.
- If applied to each pixel, execution time becomes extremely long.
- Superpixels reduce the number of processed units—from millions (e.g., 4,000,000 pixels) to only a few hundred.

Visual Saliency

- **Concept of Visual Saliency**

  - The human eye provides a large amount of information to the brain, more than the brain can fully process.

  - Thus, the brain gives priority to visually important content.

Visual Saliency

- The regions the brain processes are those that are **distinct or salient**.
- These regions differ in color or illumination, or may include fast movement (e.g., seeing a cockroach suddenly run past you very fast).
- Saliency expresses how distinct a region, person, or pixel is compared to its surroundings.
- To model saliency, **saliency maps** were introduced—ordered topographic maps representing scene saliency.
- Despite precise definitions in neuroscience and psychology, computer vision lacks a strictly defined saliency concept.
- Saliency maps can be used to predict eye fixation points—i.e., where a human is likely to look.

Visual Saliency

- Some models aim to detect salient objects or regions.
- Saliency in computer vision is part of the broader field:
  - **Computational Models for Visual Attention**, which describes mechanisms for computing attention and includes theoretical studies of brain and behavioral patterns.
  - A salient object is defined as an intrinsic property of the image—information within it—that can be clearly and explicitly perceived and understood by humans.

Visual Saliency



Visual Saliency

- The whiter a pixel is, the more important it is The blacker it is, the less important it is.
- A saliency map represents visual saliency in a scene—values in the map represent clarity and perceptual importance of regions.
- Saliency maps have large applications such as:
  - Predicting where humans will look
  - Directing cameras toward important regions
  - Predicting eye movement
  - Prioritizing important regions for transmission, generation, or compression
- Object detection, recognition, and tracking
- Segmentation

Visual Saliency

- Saliency can be determined using:
  - **Internal information only** (contrast and edges)
  - **External information** from other images.

Saliency Maps

- Algorithms vary, but the simplest is **Frequency-Tuned Saliency Detection (FT)**.

FT-Based Saliency Maps

- Saliency detection relies on global and local contrast:
  - Local contrast: difference between a pixel and its neighbors
  - Global contrast: difference between a pixel and the image’s mean color

FT Algorithm

- The FT algorithm relies on global contrast.
- It operates by:
  1. Applying a Gaussian filter
  2. Converting RGB to LAB color space
  3. Computing the average of L, A, and B channels
  4. Computing the Euclidean distance between each pixel and the average
  5. Threshold to obtain a binary image

$$T = \frac{2}{w \times h} \Sigma_i^w \Sigma_j^h I(x_i, y_j)$$

GMR Algorithm (Graph-Based Manifold Ranking)

Concept

- treat saliency as a **ranking problem on a graph** of image superpixels:
  - given a set of query nodes (background seeds or foreground seeds), rank every node by its relevance to those queries using **manifold ranking**.
  - They propose a two-stage pipeline:

    1. Rank using boundary priors (four sides individually → combine) to get an initial saliency map that suppresses background

    2. Binarize that map to select foreground queries and re-rank for the final saliency

How GMR Works

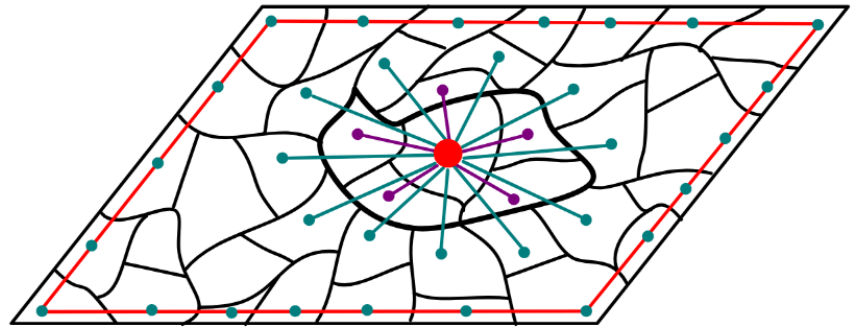
Preprocess

- Compute SLIC superpixels on `I`, producing `N` superpixels. Record for each superpixel `i`:
  - Mean color vector c_i in CIE-LAB.
  - Pixel indices to map back to full resolution.
- They default `N=200`

How GMR Works

Step 1

1. Create a graph G=(V,E)G=(V,E)G=(V,E) with `N` nodes (one per superpixel).
2. For each node i, connect:
   - its spatial neighbors (adjacent superpixels)
   - expand to `k` nearest neighbors by adjacency (k-regular extension described in paper), i.e., connect nodes that share boundaries with neighbors to capture local grouping
   - **close-loop constraint**: connect every boundary superpixel with every other boundary superpixel on the same image side.



How GMR Works

Step 1

3. Compute affinity matrix W (sparse), elementwise:

$$w_{i,j} = exp(-\frac{||c_i - c_j||^2}{\sigma^2})$$

- Only for connected pairs (others = 0). Use $\sigma^2=0.1$ as default.

4. Compute degree matrix $D = \text{diag}$ where $d_{ii} = \sum_j w_{ij}$

How GMR Works

Step 2 — Precompute ranking operator

1. Set $\alpha = 1/(1+\mu)$, they suggest $\alpha = 0.99$.
2. Form matrix $M = D - \alpha W$.
3. Compute (or prepare to solve systems with) $A = M^{-1}$.
4. Set diagonal elements of `A` to zero when used to compute ranking (to avoid self-relevance bias). (They compute inverse first then set diagonal of `A` to 0 for ranking)

How GMR Works

Step 3 — Ranking with boundary (background) queries

- For each side $S \in \{top, bottom, left, right\}$
  1. Create indicator vector y_S of length N: $y_S[i] = 1$ if superpixel i is on that image side (the paper uses nodes on that side as background queries), else 0.
  2. - Compute ranking vector $f_S = Ay_S$.
     3. Normalize `f_S` to $[0, 1]$.
  3. Convert to a saliency map for that side:

$$S - S(i) = 1 - f_s(i)$$

How GMR Works

Step 3 — Ranking with boundary (background) queries

- Combine the four side maps element-wise by multiplication (the SC rule):

$$S_{bg} = S_{top} S_{bottom} S_{left} S_{right}$$

How GMR Works

Step 3 — Ranking with foreground queries

1. Binarize S_{bq} to pick foreground seeds
2. Form indicator $y_fg[i] = 1$ for selected foreground superpixels, else 0.
3. Compute ranking $f_{fg} = Ay_{fg}$.
4. Normalize f_{fg} to $[0, 1]$. This is the **final** per-superpixel saliency:

$$S_{final}(i) = f_{fg}(i)$$